

# MAINTENANCE OPERATIONS CLIENT

## Data Model & Dax Measures

Power BI Solution | Portfolio Reconstruction

| Document Control | Detail                                                                        |
|------------------|-------------------------------------------------------------------------------|
| Engagement       | Aug-Nov 2024                                                                  |
| Solution         | Power BI dashboard built from scratch                                         |
| Data Scope       | Approximately 10,000 historical records across 20 teams                       |
| Purpose          | Define a reproducible portfolio model and KPI measure layer                   |
| Status           | Reconstructed technical specification; not the exact historical schema or DAX |

**Important:** Exact historical table structure, relationships, SQL schema and DAX expressions are not retained. The model and code below are a portfolio reconstruction based on the confirmed fields, KPI Catalogue and Power BI requirements.

## 1. Proposed Analytical Model

Use a simple star-style Power BI model with maintenance activity at the center and business dimensions around it. This keeps KPI calculations consistent and supports filtering by team, technician, priority, category, asset/equipment, location and date.

| Table           | Type      | Purpose                              | Key Fields                                                                                                                      |
|-----------------|-----------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| FactMaintenance | Fact      | Maintenance task/work-order records  | Work Order ID, Team, Technician, Asset/Equipment, Created Date, Due Date, Completion Date, Status, Priority, Category, Location |
| DimTeam         | Dimension | Standardized team filtering          | Team                                                                                                                            |
| DimTechnician   | Dimension | Technician filtering                 | Technician                                                                                                                      |
| DimAsset        | Dimension | Asset/equipment filtering            | Asset / Equipment                                                                                                               |
| DimStatus       | Dimension | Controlled status filtering          | Status                                                                                                                          |
| DimPriority     | Dimension | Priority filtering                   | Priority                                                                                                                        |
| DimCategory     | Dimension | Maintenance-category filtering       | Category                                                                                                                        |
| DimLocation     | Dimension | Location filtering where appropriate | Location                                                                                                                        |
| DimDate         | Dimension | Date-based analysis                  | Date                                                                                                                            |

**Model boundary:** These table names and the star-style structure are proposed for the portfolio. They are not claimed as the original client model.

## 2. Proposed Relationships

| From          | To              | Relationship | Purpose                                                             |
|---------------|-----------------|--------------|---------------------------------------------------------------------|
| DimTeam       | FactMaintenance | 1 : many     | Team analysis                                                       |
| DimTechnician | FactMaintenance | 1 : many     | Technician analysis                                                 |
| DimAsset      | FactMaintenance | 1 : many     | Asset/equipment analysis                                            |
| DimStatus     | FactMaintenance | 1 : many     | Status analysis                                                     |
| DimPriority   | FactMaintenance | 1 : many     | Priority analysis                                                   |
| DimCategory   | FactMaintenance | 1 : many     | Category analysis                                                   |
| DimLocation   | FactMaintenance | 1 : many     | Location analysis                                                   |
| DimDate       | FactMaintenance | 1 : many     | Date analysis; active date relationship must be explicitly selected |

**Date design:** Created Date, Due Date and Completion Date serve different business purposes. The reconstructed model should not assume all three can be active through one date relationship simultaneously.

## 3. Measure Design Rules

| Rule                   | Requirement                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------|
| Single source of truth | Measures must follow the KPI Catalogue.                                                                         |
| Completion             | Use the confirmed completion formula.                                                                           |
| Overdue                | Use the confirmed Due Date / Status rule.                                                                       |
| Status mapping         | Completed/Open/Pending depend on the documented reconstructed status mapping.                                   |
| Backlog                | Do not create a final backlog measure until one reconstructed backlog definition is approved for the portfolio. |
| Technician performance | Use individual operational measures; do not invent a composite score.                                           |
| Illustrative data      | Synthetic dashboard values must not be presented as client facts.                                               |

## 4. Core DAX Measures

**Code status:** The DAX below is reconstructed implementation code for the portfolio. Only the underlying Completion Rate and Overdue business rules are confirmed historical rules.

### 4.1 Total Assigned Maintenance Tasks

```
Total Assigned Maintenance Tasks =  
COUNTROWS ( FactMaintenance )
```

Portfolio assumption: one eligible prepared fact row represents one assigned maintenance task/work order. If the rebuilt dataset has a different grain, this measure must be adjusted.

### 4.2 Completed Maintenance Tasks

```
Completed Maintenance Tasks =  
CALCULATE (  
    [Total Assigned Maintenance Tasks],  
    DimStatus[Status] = "Completed"  
)
```

Requires the reconstructed status mapping to standardize completed records to the controlled Completed category.

### 4.3 Completion Rate

```
Completion Rate =  
DIVIDE (  
    [Completed Maintenance Tasks],  
    [Total Assigned Maintenance Tasks]  
)
```

**Confirmed definition:** Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100. Format the DAX measure as a percentage.

### 4.4 Overdue Work Orders

```
Overdue Work Orders =  
CALCULATE (  
    [Total Assigned Maintenance Tasks],  
    FILTER (  
        FactMaintenance,  
        NOT ISBLANK ( FactMaintenance[Due Date] )  
        && FactMaintenance[Due Date] < TODAY ()  
        && FactMaintenance[Status] <> "Completed"  
    )  
)
```

**Confirmed rule:** Due Date < Current Date AND Status != Completed.

### 4.5 Open Work Orders

```
Open Work Orders =  
CALCULATE (  
    [Total Assigned Maintenance Tasks],  
    DimStatus[Status] = "Open"  
)
```

Reconstructed measure. Exact historical Open-status mapping is not retained.

### 4.6 Pending Tasks

```
Pending Tasks =  
CALCULATE (  
    [Total Assigned Maintenance Tasks],  
    DimStatus[Status] = "Pending"  
)
```

Reconstructed measure. Open and Pending must remain distinct unless the rebuilt source mapping establishes equivalence.

## 5. Analytical DAX Measures

### 5.1 Average Completion Time (Days)

```
Average Completion Time (Days) =
AVERAGEX (
    FILTER (
        FactMaintenance,
        FactMaintenance[Status] = "Completed"
        && NOT ISBLANK ( FactMaintenance[Created Date] )
        && NOT ISBLANK ( FactMaintenance[Completion Date] )
        && FactMaintenance[Completion Date] >= FactMaintenance[Created Date]
    ),
    DATEDIFF (
        FactMaintenance[Created Date],
        FactMaintenance[Completion Date],
        DAY
    )
)
```

**Reconstruction boundary:** Created Date is used here as the portfolio start/reference date. The exact historical start-date convention is not retained.

### 5.2 Team Completion Rate

```
Team Completion Rate =
[Completion Rate]
```

No separate formula is required. The Completion Rate measure naturally recalculates within Team filter context.

### 5.3 Technician Completion Rate

```
Technician Completion Rate =
[Completion Rate]
```

Use within Technician filter context; missing technician attribution must remain transparent.

### 5.4 High-Priority Pending Work

```
High-Priority Pending Work =
CALCULATE (
    [Total Assigned Maintenance Tasks],
    DimPriority[Priority] = "High",
    DimStatus[Status] = "Pending"
)
```

**Reconstruction boundary:** The exact historical priority vocabulary is not retained. 'High' is a portfolio mapping label and must match the sanitized reconstructed dataset.

## 6. Backlog Measure Boundary

Backlog was a confirmed analytical finding, but the exact historical backlog definition is not retained. A final DAX formula should therefore be created only after the portfolio explicitly defines which outstanding records qualify as backlog.

| Required Decision              | Why It Must Be Defined                                                      |
|--------------------------------|-----------------------------------------------------------------------------|
| Eligible statuses              | Determines which outstanding records enter backlog.                         |
| Age/due-date condition, if any | Determines whether backlog includes all outstanding work or only aged work. |
| Treatment of overdue           | Prevents backlog and overdue from being silently treated as identical.      |
| Missing due dates              | Defines whether those records can enter backlog.                            |
| Dataset grain                  | Prevents duplicate/inflated backlog counts.                                 |

**Do not fabricate:** No backlog DAX is presented as historical or final until this reconstructed business rule is explicitly defined.

## 7. Recommended Measure Set by Dashboard Area

| Dashboard Area            | Measures                                                                                                                   |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Executive Overview        | Total Assigned Maintenance Tasks, Completed Maintenance Tasks, Completion Rate, Overdue Work Orders, reconstructed Backlog |
| Work Order & Completion   | Completed, Open, Pending, Overdue, Average Completion Time                                                                 |
| Team & Technician         | Completion Rate in Team/Technician context, assigned workload, completed work, overdue work                                |
| Backlog, Priority & Asset | Reconstructed Backlog, High-Priority Pending, Overdue, counts by Category and Asset/Equipment                              |

## 8. Validation Checks

| ID     | Validation                                                                                        |
|--------|---------------------------------------------------------------------------------------------------|
| DAX-01 | Completion Rate reconciles to Completed Maintenance Tasks / Total Assigned Maintenance Tasks.     |
| DAX-02 | 92% is presented only as the retained overall engagement result, not hard-coded into the measure. |
| DAX-03 | Every overdue record has Due Date < Current Date and Status != Completed.                         |
| DAX-04 | Completed records do not appear in Overdue Work Orders.                                           |
| DAX-05 | Open and Pending measures follow the documented reconstructed status mapping.                     |
| DAX-06 | Average Completion Time excludes missing/invalid date intervals under the rebuilt rule.           |
| DAX-07 | Team and technician views reuse governed base measures rather than redefining KPI logic.          |
| DAX-08 | Backlog uses one explicit reconstructed definition before implementation.                         |
| DAX-09 | Equivalent measures reconcile across dashboard pages under equivalent filters.                    |

## 9. Traceability

| Measure                    | KPI Catalogue  | Power BI Requirement |
|----------------------------|----------------|----------------------|
| Total Assigned / Completed | KPI-01, KPI-02 | PBI-01, PBI-02       |
| Completion Rate            | KPI-01         | PBI-01               |
| Open / Pending             | KPI-03, KPI-04 | PBI-02               |
| Overdue Work Orders        | KPI-05         | PBI-03               |
| Average Completion Time    | KPI-06         | PBI-04               |
| Team Completion            | KPI-08         | PBI-06               |
| Technician Completion      | KPI-09         | PBI-07               |
| Backlog                    | KPI-10         | PBI-08               |
| High-Priority Pending      | KPI-11         | PBI-05               |

## 10. Implementation Notes

This document provides enough technical detail to rebuild a defensible portfolio Power BI model without pretending that the original client schema or DAX is still available. The model should be implemented against sanitized or synthetic portfolio data and tested against the KPI definitions before dashboard screenshots are produced.

**Public portfolio:** Any screenshot using synthetic values must be labeled: SANITIZED PORTFOLIO MOCKUP - ILLUSTRATIVE VALUES, NOT CLIENT DATA.